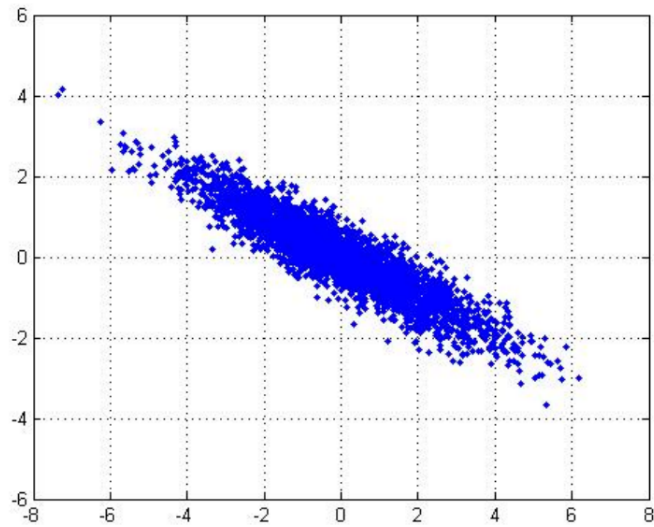


Modélisation mathématique

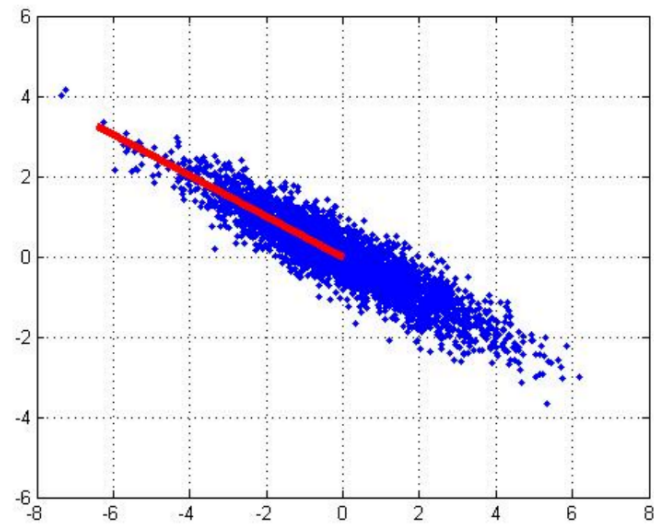
Module R5.12

Argheesh Bhanot

ACP



ACP



ACP

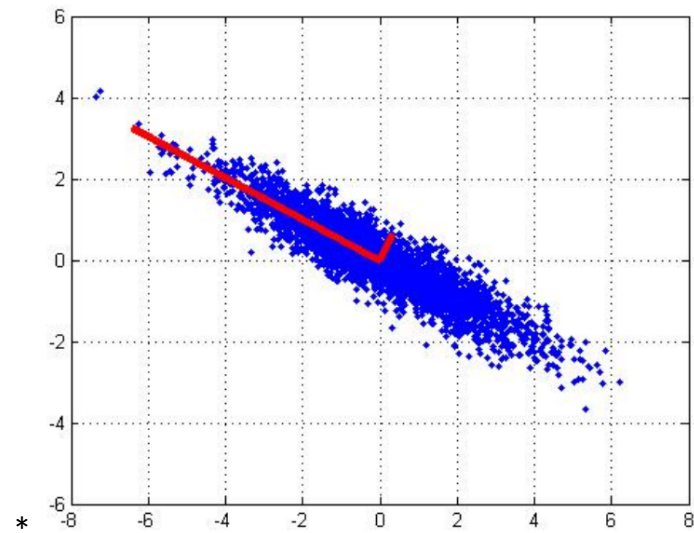
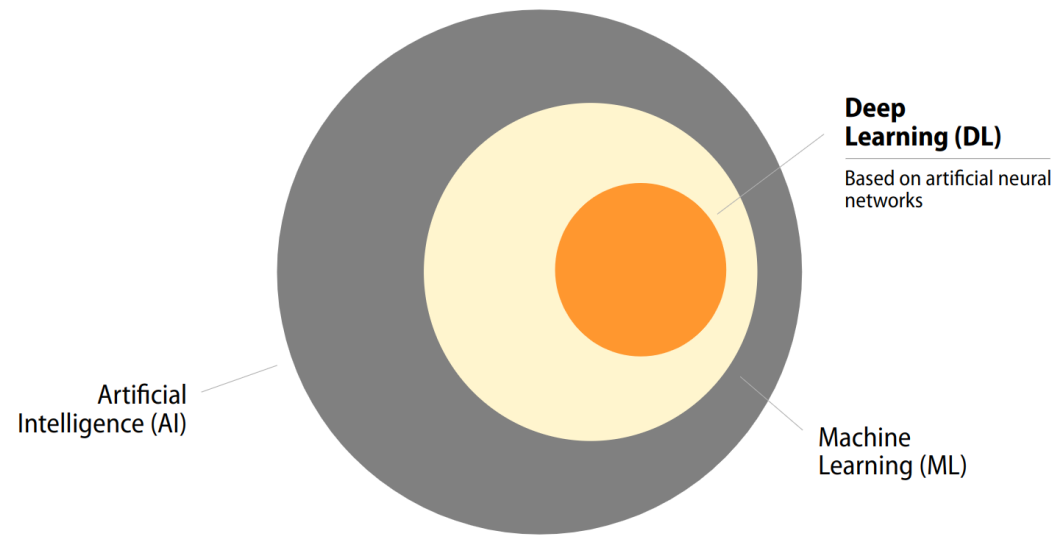


Diagramme de Venn d'IA



Apprentissage par machine
●○○○○○○○○○

1 Apprentissage par machine

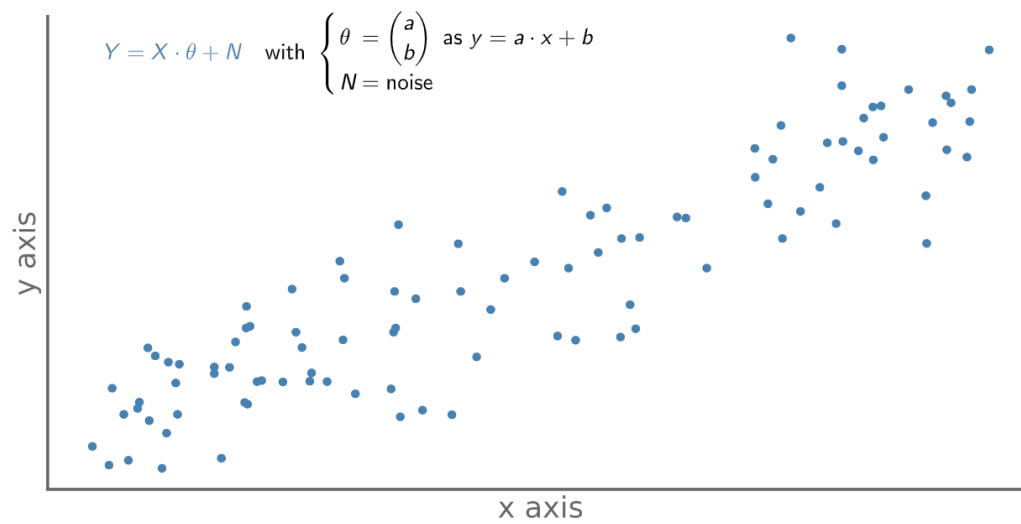
Apprentissage par machine

Un algorithme qui arrive à apprendre des données

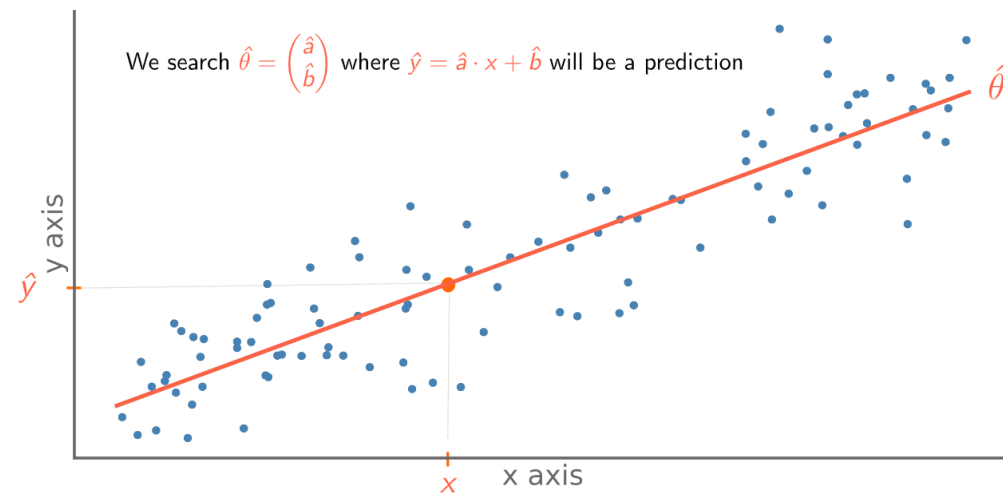
Définition : Taches, experiences, mesure de performance (Mitchell 1997)

A computer program is said to learn from experience (E) with respect to some class of tasks(T) and performance measure(P), if its performance at tasks in(T), as measured by (P), improves with experience (E).

Régression linéaire



Régression linéaire



Régression linéaire

For this, we will use an «loss function», which we will try to minimize.

$$\text{RMSE}(X, h_{\theta}) = \sqrt{\frac{1}{n} \sum_{i=1}^n [h_{\theta}(X^{(i)}) - Y^{(i)}]^2}$$

$$\text{MSE}(X, h_{\theta}) = \frac{1}{n} \sum_{i=1}^n [h_{\theta}(X^{(i)}) - Y^{(i)}]^2$$

